

Chapter 1: Physical Quantities, Units and Measurements

1.1 Physical Quantities

Physical Quantity (definition)

- A physical quantity is a quantity that can be measured.
- It consists of a magnitude and a unit.

- Physical quantities can be grouped into two types:
Base quantities and Derived quantities.

- Each of these quantities has an international system (SI) unit.

- Example: $100\text{ m} \rightarrow$ Unit
 $\hookrightarrow$ Magnitude

Base quantities

- There are 7 base quantities. (Need to know 6 for O levels)

Base quantity	SI unit	Symbol of SI unit
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electrical current	ampere	A
Amount of substance	mole	mol
Luminous Intensity (not in O-levels)	candela	cd

Derived quantities

- These physical quantities can be derived from the 7 base quantities.

Physical quantity	How it is derived	Symbol of unit
Area	length $\times$ width	m^2
Speed	length $\div$ time	m/s

Expressing units using the base units

Example: pascal (Pa)

Pressure = Force / Area

Unit of force in terms of base unit = kg m/s^2

Unit of area in terms of base unit = m^2

$$\therefore \text{Pressure} = \frac{\text{kg m/s}^2}{\text{m}^2}$$

$$= \frac{\text{kg m}}{\text{s}^2} \div \text{m}^2$$

$$= \frac{\text{kg m}}{\text{s}^2} \times \frac{1}{\text{m}^2}$$

$$= \frac{\text{kg}}{\text{m s}^2}$$

$$= \text{kg/(m s}^2\text{)}$$

Use the bracket to indicate the denominator!!

Expressing the magnitude of physical quantities

1. Using standard form

Standard form = $A \times 10^b$ where $1 \leq A < 10$ and b is integer.

Example: $640000 = 6.4 \times 10^5$

How to convert:

- Find the decimal point.
- Keep moving the decimal point left or right until $1 \leq A < 10$.
- Every time you move left, $b+1$.
- Every time you move right, $b-1$.
- (b starts at 0.)

2. Using prefix

Prefix	Symbol	Factor
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

Examples:

1m = 10dm = 100cm = 1000mm → Use calculator

1 μ m = 10^{-6} m ENG function

★ MUST KNOW: Thickness of hair → 0.1mm

Diameter of atom → 0.1nm

Diameter of earth → 10000km

1.2 Measurement

Measurement range of instrument

- Example: 0mm to 25mm, -10°C to 110°C

↳ Upper & Lower Limits of an instrument

Precision of instrument

- The precision of an instrument is the smallest non-zero value that the instrument can measure.

Measurement errors

- Two main types of errors

- Random errors → keeps fluctuating, cannot get rid of
- Systematic errors → instrument problem

	Random Errors	Systematic Errors
What it is	- Random magnitude	- Constant magnitude
Occurrence	- In all measurements. - Observers estimate the last figure of a reading on an instrument.	- Observers consistently overestimate or underestimate a reading.
How to get rid	- Reduced by averaging a large number of readings.	- Minimised by method of differences.

Random Error - Parallax Error

- Takes place due to the positioning of the eye when reading the ^{measurement} ~~error~~ from the measuring instrument.
- To prevent parallax error, the eye should be placed such that the line of sight is directly above the scale marking to read.

System Error - Zero Error

- Takes place when the metal in the caliper gets deformed over time/rusty. (wear & tear)
- ALWAYS PRESS THE ZERO BUTTON to reset the display to zero.

Measurement of Length

- Common instruments to measure length:

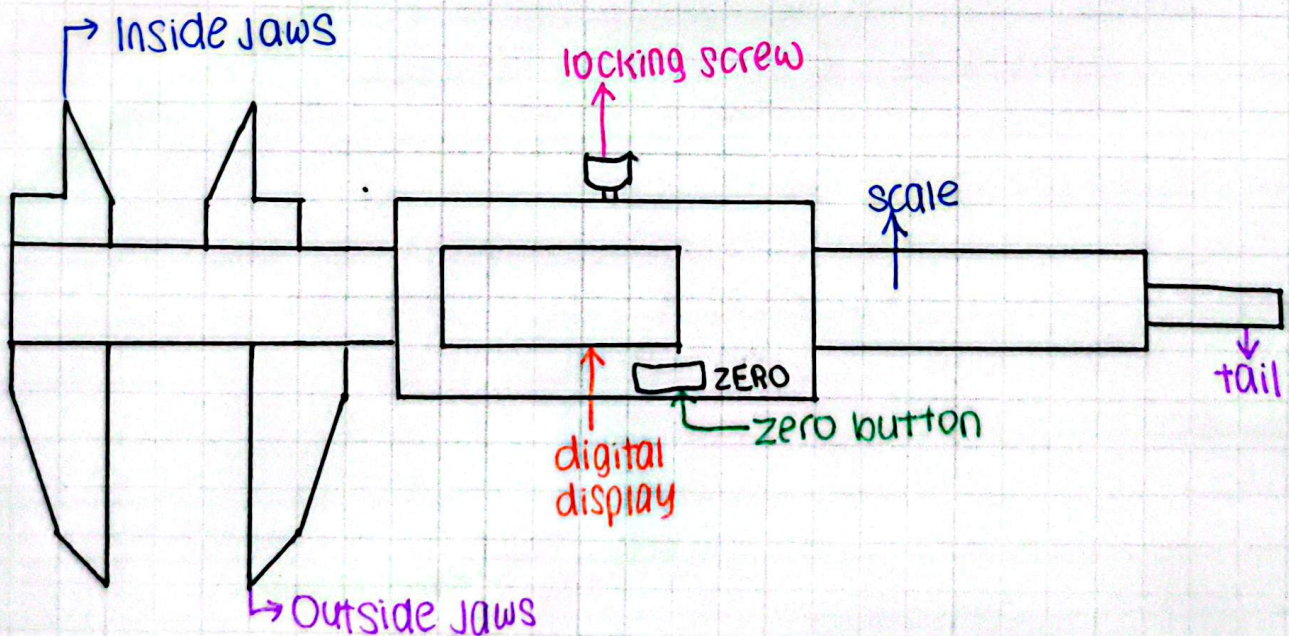
→ Metre rule, tape measure, digital calipers, digital micrometer

Screw gauge

Instrument	Measuring range	Precision
Metre rule	0-1m	0.1cm or 1mm
Tape measure	0-5m	0.1cm or 1mm
Digital calipers	0-15cm	0.01cm or 0.1mm
Digital micrometer	0-2.5cm	0.001cm or 0.01mm
Screw gauge		

Digital calipers

- An instrument for measuring the internal and external diameters of cylinders or diameter of circular objects or depth of a container.



1. Inside Jaws

→ Used to measure internal diameter of object

2. Outside Jaws

→ used to measure external diameter or width of object

3. Locking Screw

→ used to ensure that the jaws do not move apart

4. Digital Display

→ used to show the numeric value

5. Zero Button

→ used to reset the displayed length back to zero

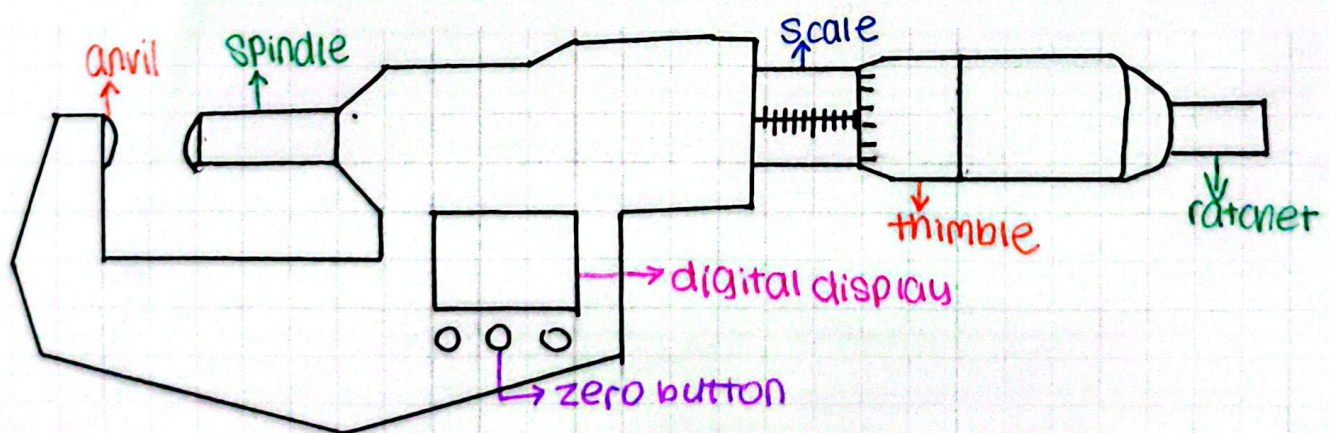
6. Scale

7. Tail

→ used to measure the depth of an object.

Digital Micrometer Screw Gauge

- A useful instrument to measure external diameters or thickness of small objects.



1. Anvil and spindle

→ used to measure the diameter of an object

2. Zero Button

→ Used to reset the displayed length back to zero.

3. Digital Display

→ Used to show the numeric value

4. Ratchet

→ Can be turned to adjust the spindle so that the spindle is in contact with the object

Measurement of Volume

- Common measuring instruments for volume:

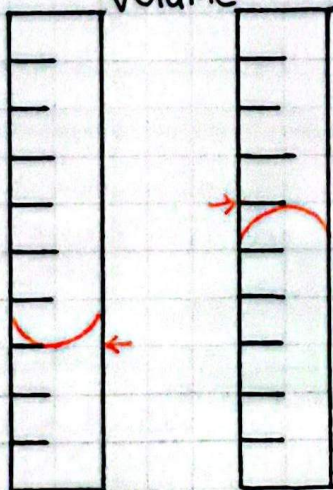
→ measuring cylinder, volumetric flask, burette, pipette

How to measure volume

- Ensure that the cylinder is set up in a vertical position.

- The eye is at the same level as the meniscus (curved surface) of the liquid.

How to measure volume



Measurement of Time

- Stopwatches are used to measure short intervals of time.

Types: Digital stopwatch, analogue stopwatch

Stopwatch

- Used to measure short intervals of time

Digital stopwatch $\rightarrow$ more accurate as it measures time in intervals of 0.01s.

Analogue stopwatch $\rightarrow$ measures time in intervals of 0.1s.

Pendulum

- A simple pendulum consists of a bob attached to a string.

* Length $\rightarrow$ Point of suspension to centre of bob

Complete oscillation (definition)

- One complete oscillation is a complete to-and-fro motion (from $A \rightarrow B \rightarrow A$ or $O \rightarrow B \rightarrow A \rightarrow O$)

Period (definition)

- The period of a ~~simple~~ simple pendulum is the time taken for one complete oscillation.

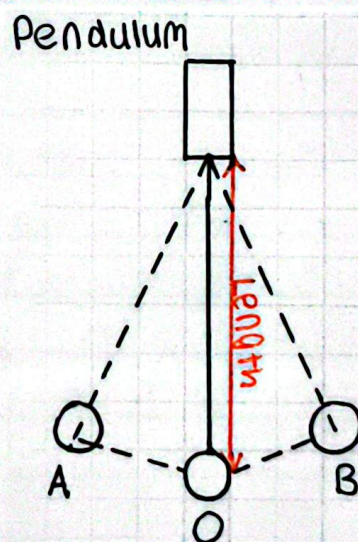
- When measuring period of pendulum, swing for about 20 oscillation instead of 1. This minimises error caused by human reaction time. Repeat a few times (and average it) for better accuracy.

* The period is dependent on the length of the pendulum.

The mass of the bob DOES NOT CHANGE PERIOD.

- When the length of the pendulum increases, period increases.

- When the length of the pendulum decreases, period decreases.



Errors in Measuring Time

- Do not round time to nearest 0.01s because of human reaction time in starting and stopping the watch.
- Always take more than one reading and take the average.

Doing calculation in physics

- When adding/subtracting values of different decimal place, use the smaller decimal place. e.g. $3.55g + 4.1g = 7.7g$
- Same for significant figures, choose the smallest number of significant figure.

1.3 Scalars and Vectors

Scalar Quantities (definition)

- Scalar quantities are ~~quantities~~ physical quantities that have only magnitude.

Vector Quantities (definition)

- Vector quantities are physical quantities that have both magnitude and direction.

Examples of scalar and vector Quantities

Scalar	Vector
Distance	Displacement
Speed	Velocity
Mass	Acceleration
Energy	Force
Time	Weight

Vector Diagrams (Addition of vectors)

- A vector diagram is used to add up two or more vectors.
- A vector quantity is represented by an arrow, where the length of arrow is proportional to magnitude of vector; direction of arrow indicates direction of vector
- A resultant vector is produced, and it must be equivalent to the individual vectors combined in terms of magnitude and direction.

→ Vector

→→ Resultant Vector

Adding parallel vectors

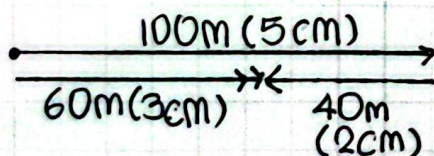
Example: If a girl walks 100m due east then 40m due west, what is her displacement?

Step 1: Choose an appropriate scale.

Scale: 1cm to 20m

Step 2: Label starting point.

Step 3: Draw the arrows.



Step 4: Formulate answer:

(quantity)(distance)(direction)

The displacement of the girl is 60m due East / to the right of starting point.

Adding non-parallel vectors

- Parallelogram Method
- Tip-to-tail Method
- Measure the resultant vector (length & angle)

- Format to write:

→ The resultant (quantity) has a magnitude of ____ + (unit) and acts at an angle (angle) (clockwise/anticlockwise) from (reference line).

Parallelogram Method

Example: There are two non-parallel forces acting on a block, 4N and 6N. Find the resultant force (magnitude and direction)

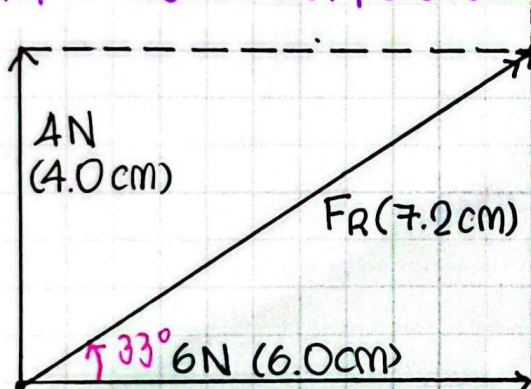
Step 1: Choose an appropriate scale

Scale: 1.0cm to 1.0N

Step 2: Draw the forces coming out from a COMMON POINT.

↳ Rearrange if it is not.

Step 3: Construct parallelogram



Step 4: Draw resultant force

Step 5: Determine angle & length of resultant force.

Angle $\rightarrow 33^\circ$
Resultant force $\rightarrow 7.2\text{cm} \times 1.0$
 $= 7.2\text{N}$

The resultant force has magnitude of 7.2N and acts at an angle 33° anti-clockwise from the 6N force.

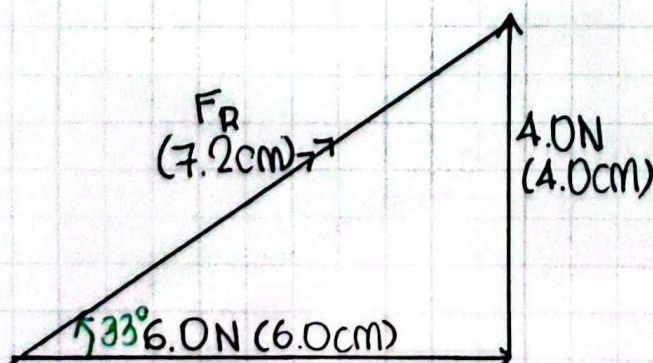
Tip-to-tail Method

Step 1: Choose an appropriate scale

Scale: 1.0cm to 1.0N

Step 2: Draw one arrow first.

Step 3: Then, join the other arrow such that the head of the first arrow touches the tail of the other arrow.



Step 4: Connect the arrows, forming a triangle.

Step 5: Determine length & angle of resultant force.

Angle $\rightarrow 33^\circ$
Resultant Force $\rightarrow 7.2\text{N}$

The resultant force has a magnitude of 7.2N and acts at an angle 33° anti-clockwise from 6N force.